

Task 2 – Web Vulnerability Testing Report

Objective

The objective of this task is to identify common web vulnerabilities such as Cross-Site Scripting (XSS) and SQL Injection (SQLi) using a safe demo application environment. The task also involves intercepting HTTP requests using Burp Suite and suggesting mitigation techniques.

Tools Used

- Burp Suite
 - DVWA
 - Web Browser
 - Kali Linux
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Introduction

Web applications are common targets for cyberattacks. Vulnerabilities such as SQL Injection and Cross-Site Scripting (XSS) can allow attackers to steal sensitive data, manipulate databases, or execute malicious scripts.

In this task, a safe testing environment was used to identify basic vulnerabilities and understand secure coding practices.

Step 1 – SQL Injection Testing

Objective

To test whether user input fields are vulnerable to SQL Injection attacks.

Test Payload Used

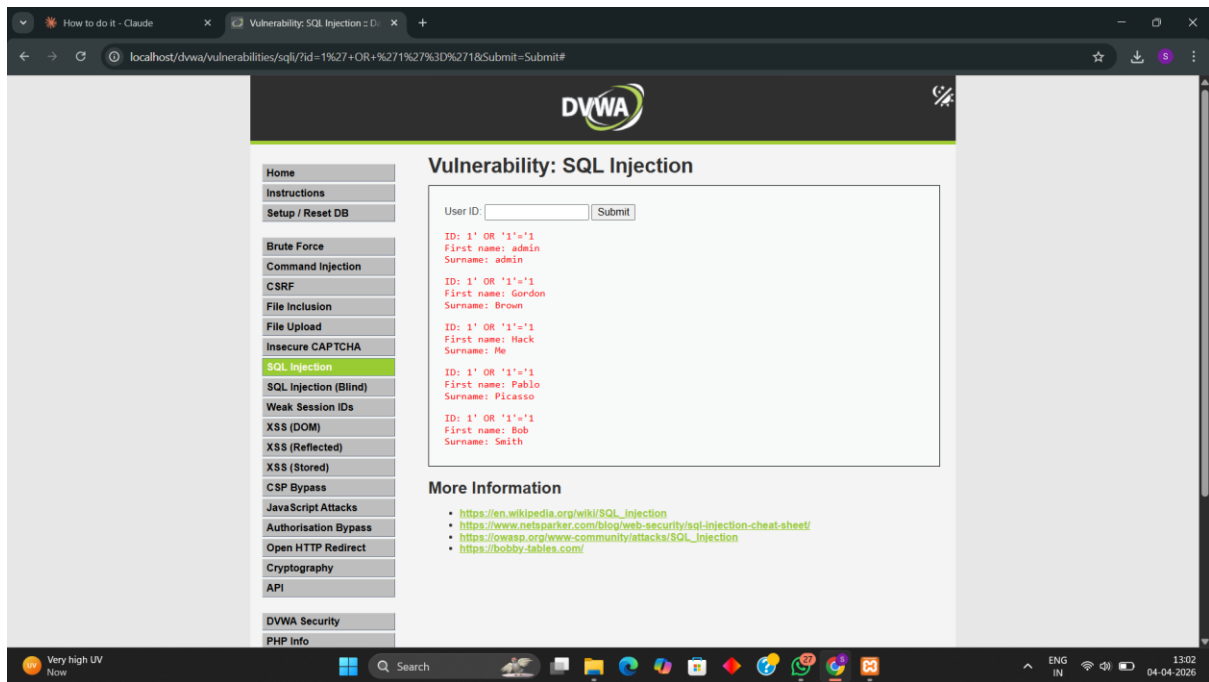
' OR '1'='1

Procedure

- Opened the DVWA SQL Injection module
- Entered the payload into the input field
- Observed the server response

Findings

The application accepted the malicious SQL query and returned unintended database results, indicating improper input validation.



Risk

Attackers may:

- Bypass authentication
- Access sensitive database information
- Modify or delete records

Step 2 – Cross-Site Scripting (XSS) Testing

Objective

To test whether the application is vulnerable to script injection attacks.

Test Payload Used

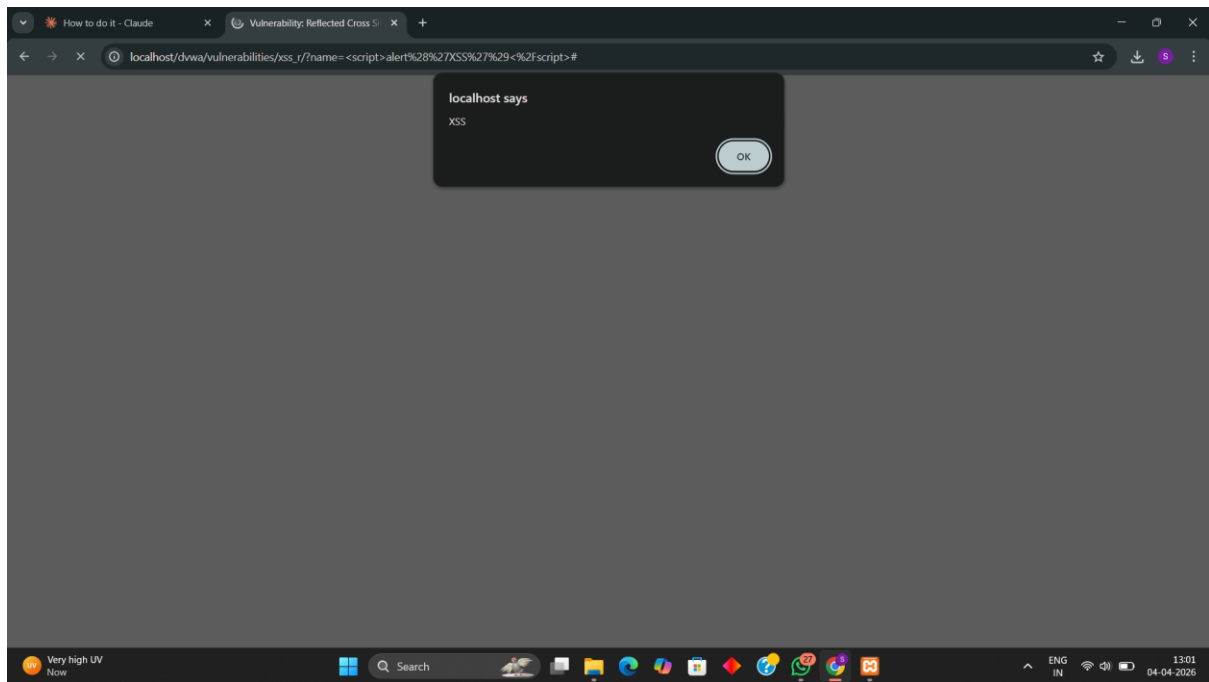
```
<script>alert('XSS')</script>
```

Procedure

- Opened the XSS module in DVWA
- Submitted the payload through the input form
- Observed JavaScript execution in the browser

Findings

The browser executed the injected script successfully, confirming the presence of an XSS vulnerability.



Risk

Attackers may:

- Steal cookies/session data
 - Redirect users to malicious websites
 - Execute malicious scripts in user browsers
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Step 3 – Burp Suite Request Interception

Objective

To intercept and inspect HTTP requests between the browser and the web application.

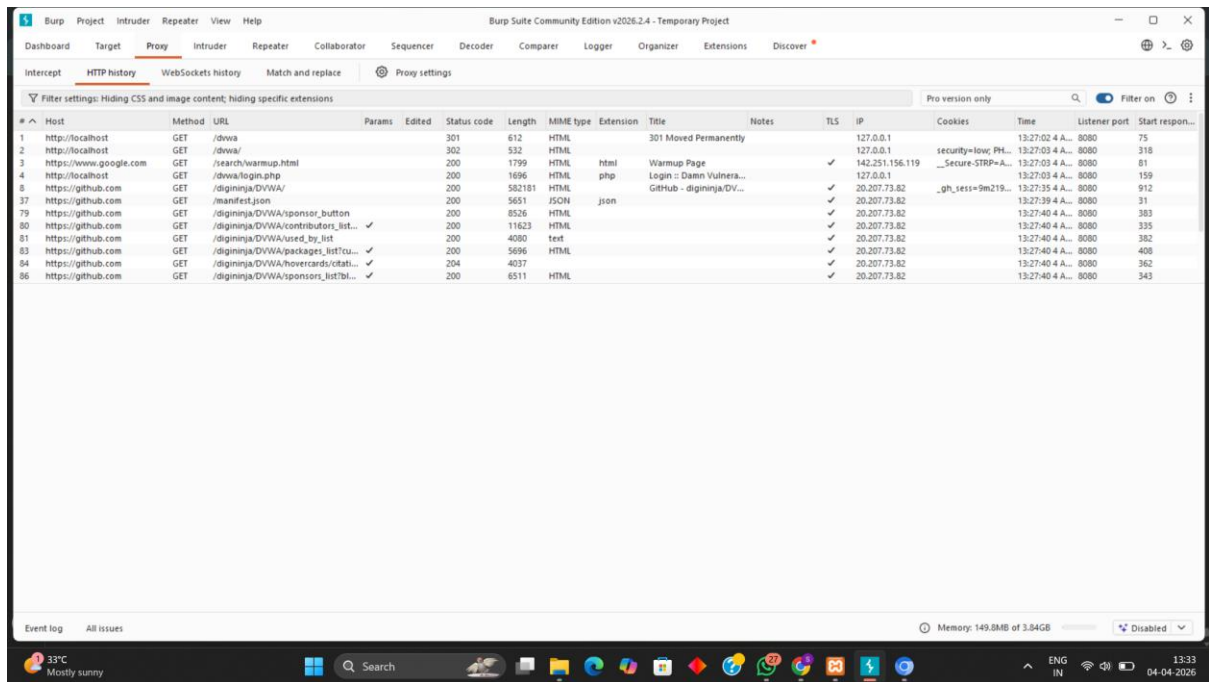
Procedure

- Configured browser proxy settings
- Enabled intercept mode in Burp Suite
- Captured HTTP GET and POST requests

Findings

Burp Suite successfully intercepted requests and displayed:

- Request headers
 - Parameters
 - Cookies
 - Server responses
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Mitigation Recommendations

Vulnerability

Recommendation

SQL Injection

Use prepared statements and parameterized queries

Cross-Site Scripting (XSS)

Validate and sanitize user inputs

Weak Input Validation

Implement server-side validation

Session Exposure

Use secure cookies and HTTPS

Information Disclosure

Hide unnecessary server details

Conclusion

This task demonstrated how common web vulnerabilities can be identified in a safe testing environment. SQL Injection and XSS vulnerabilities highlight the importance of secure coding practices, proper input validation, and secure application configuration.

Ethical Statement

All testing activities were conducted only in authorized demo environments for educational purposes.